MECHANICAL PROPERTIES OF METAL

(STRUCTURAL STEEL AND SHEET METAL WORK)

SECTION: Welding and Fabrication
SCHOOL: Government Technical College (Ikotun)
SUB-TOPIC: Strength and Brittleness

STRENGTH

Definition of Strength.

Strength is the mechanical property of a metal, which provides resistance to an external force. It is typically measured by it yield strength, ultimate tensile strength (UTS), and hardness.

Metal that possesses strength.

1. Steel
2. Aluminum
3. Titanium
4. Copper
5. Brass
6. Bronze
7. Stainless steel

BRITTLENESS

Definition of Brittleness.

Brittleness is the property of a metal that fractures when subjected to stress but has little tendency to deform before rupture.

Metal that possesses brittleness.

1. Germanium (Ge).
2. Cast iron (Fe-C).
3. Chromium (Cr).
4. Molybdenum (Mo).
5. Bismuth (Bi).
6. Antimony (Sb).
7. Cobalt (Co).
8. Manganese (Mn).
9. Tungsten (W).

NAMES: Okikiola, Habib, Blessing, etc.

SURFACE HARDENING
(STRUCTURAL STEEL AND SHEET METAL WORK)

SECTION: Welding and Fabrication
SCHOOL: Government Technical College (Ikotun)
SUB-TOPIC: Cyaniding

CYANIDING

Cyaniding in context of metal, refers to a surface treatment process known as cyanide hardening or cyanide case-hardening. It is a method used to improve the hardness and wear resistance of certain of steel.

During cyaniding process, the metal part to be treated is heated in a furnace and then immersed in a cyanide salt bath, usually containing Sodium cyanide. The high temperatures and the presence of cyanide cause carbon and hydrogen atoms to diffuse into the surface layer of the metal. This creates a hardened layer with increased wear resistance.

Cyaniding is commonly used for low-carbon and low-alloy steels where the hardened layer is desired to enhance the surface properties without affecting the core properties of the metal.

METALS THAT POSSESS CYANIDE

Cyaniding is a surface hardening process that can be applied to various steel alloys, including;

1. Low-carbon steels (ALSI 1010, 1015, 1020)
2. Medium-carbon steels (AISI 1030, 1040, 1050)
3. High-carbon steels (AISI 1060, 1070, 1080)
4. Alloy steels (AISI 4130, 4140, 8620)
5. Stainless steels (some grades, like 410, 420, 440)

Cyaniding is particularly suitable for steels with a carbon content between 0.1% and 1.2%.the process can also be applied to other method such as;

1. Cast iron.
2. Nickel alloys (e.g. Inconel, Monel)
3. Cobalt alloys (e.g. Stellite)

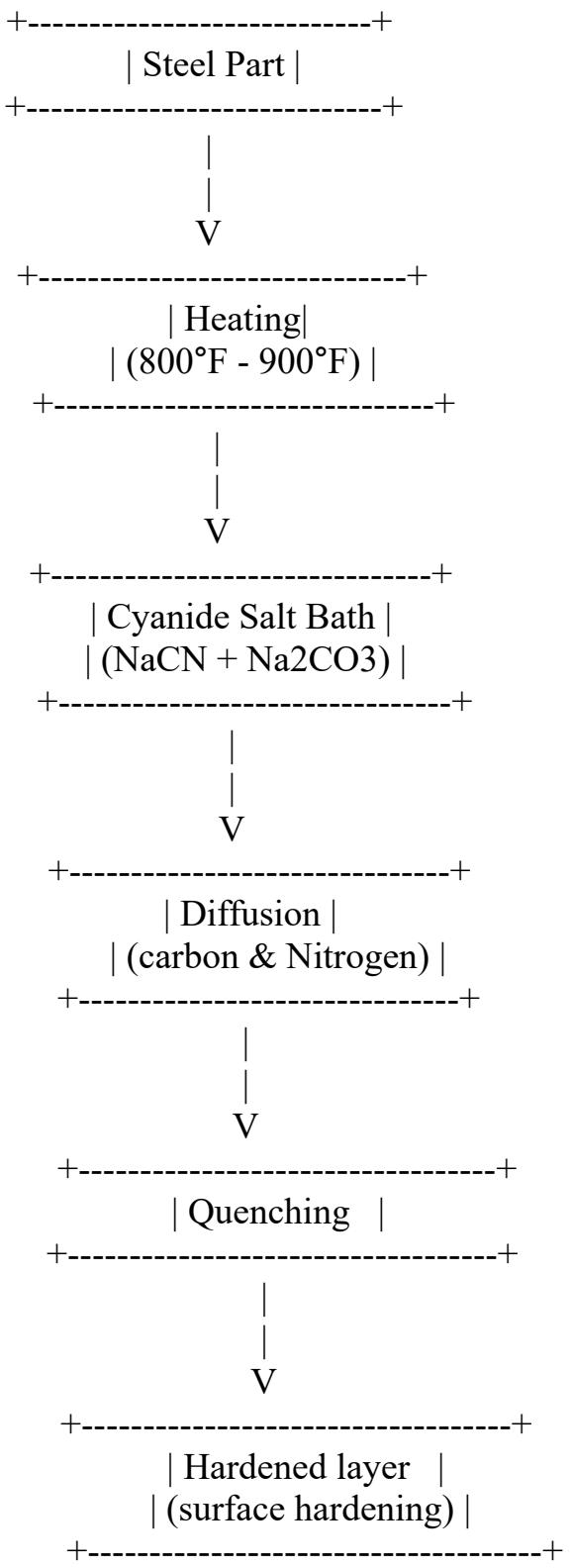
BENEFIT OF CYANIDING

1. **Surface hardening:** cyaniding creates a hard, wear resistance layer on the steel surface, improving its durability and resistance to abrasion.
2. **Improves wear resistance:** the hardening layer reduces friction and wear, extending the life span of moving part and reducing maintenance cost.
3. **Enhance fatigue strength:** cyaniding improves the steel ability to withstand repeated stress and loading reducing the risk of fatigue failure.

4. **Corrosion resistance:** the hardening layer provides some protection against corrosion especially in harsh environment.
5. **Improves machinability:** cyaniding can improve the machinability of steel part making them easier to machine and finish.

DIAGRAMMATIC REPRESENTATION OF CYANIDING IN SURFACE HARDENING

Here's a simple drawing to illustrate the cyaniding process



The drawing above shows the basic steps of the cyaniding process;

1. Heating the steel part.
2. Immersion in the cyanide salt bath.
3. Diffusion of Carbon and Nitrogen into the steel surface.
4. Quenching to completing the hardening process.
5. Formation of hardened layer on the steel surface.

ADVANTAGES OF CYANIDING IN SURFACE HARDENING

- **Rapid and efficient:** cyaniding is a fast process that can be completed in 20 to 30 minutes
- **Thin, hard shell:** cyaniding produces a thin, hard shell that is harder than the one produced by carburizing.
- **Minimal distortion:** cyaniding causes minimal distortion because the process is quick.
- **Economical:** cyaniding is a cost-effective method for surface hardening.
- **Versatility:** cyaniding can be applied to small part and tool made of low-carbon steels.

DISADVANTAGES OF CYANIDING IN SURFACE HARDENING

- **Toxicity:** cyaniding uses toxic cyanide salt which can be harmful if not handled properly.
- **Environmental concerns:** cyanide is an environmental hazard, and its disposal is strictly regulated.
- **Limited depth:** cyaniding only hardens a thin layer (typically 0.1-0.5mm) of the steel surface.
- **Not suitable for all steels:** cyaniding is only suitable for certain steel grades, and may not be effective on all type of steels.
- **Equipment corrosion:** cyanide can corrode equipment and containers, requiring specialized handling and storage.

NAMES: Okikiola, Habib, Blessing, Salami Samuel etc.